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Tutorial Sheet - 02

PART - A

Q. 1 Define linear dependent and linear independent of vectors.

Sol^n Vectors (matrices) $x_1, x_2, \ldots x_n$ are said to be dependent if

(1) All the vectors (row or column matrices) are of the same order.

(2) n scalars $\lambda_1, \lambda_2, \ldots \lambda_n$ (not all zero) exist such that

$$\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 + \ldots + \lambda_n x_n = 0$$

Other-wise they are linearly independent.

Q.2 Find the eigen values of the matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$

Sol^n Eigen value $= |A - \lambda I| = 0 \Rightarrow \begin{bmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{bmatrix}$

$$\begin{vmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{vmatrix} = 0$$

$\Rightarrow (1-\lambda)(6-2\lambda-3\lambda+\lambda^2-2) - 1(2-4+2\lambda) = 0$

$\Rightarrow -\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$

$\Rightarrow \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$\Rightarrow \lambda^3 - \lambda^2 - 5\lambda^2 + 5\lambda + 6\lambda - 6 = 0$

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$$\lambda^2(\lambda-1) - 5\lambda(\lambda-1) + 6(\lambda-1) = 0$$

$$(\lambda-1)(\lambda^2 - 5\lambda + 6) = 0$$

$$(\lambda-1)(\lambda^2 - 2\lambda - 3\lambda + 6) = 0$$

$$(\lambda-1)(\lambda(\lambda-2) - 3(\lambda-2)) = 0$$

$$(\lambda-1)(\lambda-2)(\lambda-3) = 0$$

$$\boxed{\lambda = 1, 2, 3}$$ Ans

Q.3 Define Orthogonal matrix.

Solⁿ

A square matrix A is called Orthogonal if

$$AA^T = A^TA = I$$

Q.4 Define Cayley Hamilton Theorem.

Solⁿ Statement :- Every square matrix satisfies its own characteristic equation.

If $|A-\lambda I| = (-1)^n(\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \dots + a_n)$ be the characteristic polynomial of $n\times n$ matrix $A = (a_{ij})$, then the matrix equation

$$X^n + a_1X^{n-1} + a_2X^{n-2} + \dots + a_nI = 0$$ is satisfied by $X = A$ i.e,

$$A^n + a_1A^{n-1} + a_2A^{n-2} + \dots + a_nI = 0$$

Q.5 Define model matrix.

Solⁿ A model matrix represents transformation or mapping in linear algebra applications (like rotation, scaling, projection etc.).

PART - B

Q.1 Are the vectors $x = (1,3,4,2)$, $y = (3,-5,2,2)$ and $z = (2,-1,3,2)$ linearly dependent? If so express one of these as a linear combination of others.

Sol^n

$$A = \begin{bmatrix} 1 & 3 & 4 & 2 \\ 3 & -5 & 2 & 2 \\ 2 & -1 & 3 & 2 \end{bmatrix} \qquad R_2 = R_2 - 3R_1 ,\ R_3 = R_3 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 3 & 4 & 2 \\ 0 & -14 & -10 & -4 \\ 0 & -7 & -5 & -2 \end{bmatrix} \qquad R_3 = 2R_3 - R_2$$

$$\begin{bmatrix} 1 & 3 & 4 & 2 \\ 0 & -14 & -10 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \rho(A) = 2$$

$\{ 2 < 3 = \text{no. of vectors} \}$

Vectors of are linearly dependent.

For Sol^n, x_1, x_2, x_3

$$x_2 = \lambda_1 x_1 + \lambda_2 x_3$$

$$(3,-5,2,2) = \lambda_1 (1,3,4,2) + \lambda_2 (2,-1,3,2)$$

$3 = \lambda_1 + 2\lambda_2$ —— (1)

$-5 = 3\lambda_1 - \lambda_2$

$2 = 4\lambda_1 + 3\lambda_2$

$2 = 2\lambda_1 + 2\lambda_2$ —— (2)

Eleminating eq^n (2) with (1)

$2\lambda_1 + \cancel{2\lambda_2} = 2$

$\lambda_1 + \cancel{2\lambda_2} = 3$

(−) (−) (−)

$\lambda_1 = -1$

Put $\lambda_1 = -1$ in eq. (1)

$-1 + 2\lambda_2 = 3$

$2\lambda_2 = 4$

$\lambda_2 = 2$

$\lambda_1 = -1$, $\lambda_2 = 2$ Ans

Q.2 Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$.

Sol^n $|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix} = 0$

$(8-\lambda)(21 - 7\lambda - 3\lambda + \lambda^2 + 16) + 6(-18 + 6\lambda + 8) + 2(24 - 14 + 2\lambda) = 0$

$\Rightarrow \lambda^3 - 18\lambda^2 + 45\lambda = 0$

$\lambda(\lambda^2 - 18\lambda + 45) = 0$

$\lambda(\lambda - 3)(\lambda - 15) = 0 \Longrightarrow \lambda = 0, 3, 5$ {Eigen Values}

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Now Eigen Vector, let x be an eigen vector, $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ & $(A-\lambda I)X = 0$

$$\begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{—— } (1)$$

Put $\lambda = 0$ in eqn (1), then

$$8x_1 - 6x_2 + 2x_3 = 0$$
$$-6x_1 + 7x_2 - 4x_3 = 0$$
$$2x_1 - 4x_2 + 3x_3 = 0$$

Using Cross Multiplication Method :-

$$a_1 x_1 + b_1 x_2 + c_1 x_3 = 0$$
$$a_2 x_1 + b_2 x_2 + c_2 x_3 = 0$$

$$\frac{x_1}{24-14} = \frac{-x_2}{-32+12} = \frac{x_3}{56-36} = K_1 \quad \text{or} \quad \frac{x_1}{10} = \frac{-x_2}{-20} = \frac{x_3}{20} = K_1$$

or $\frac{x_1}{1} = \frac{x_2}{2} = \frac{x_3}{2} = K_1$

$$x_1 = K_1 \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Put $\lambda = 3$ in eqn (1), then

$$5x_1 - 6x_2 + 2x_3 = 0$$
$$-6x_1 + 4x_2 - 4x_3 = 0$$
$$2x_1 - 4x_2 = 0$$

$$\frac{x_1}{24-8} = \frac{-x_2}{-20+12} = \frac{x_3}{20-36} = K_2 \quad \text{or} \quad \frac{x_1}{16} = \frac{-x_2}{-8} = \frac{x_3}{-16} = K_2$$

or $\frac{x_1}{2} = \frac{x_2}{1} = \frac{x_3}{-2} = K_1$

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$$x_1 = K_2 \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

Put $\lambda = 15$ in eq^n (1), then

$$-7x_1 - 6x_2 + 2x_3 = 0$$
$$-6x_1 - 8x_2 - 4x_3 = 0$$
$$2x_1 - 4x_2 - 12\lambda_3 = 0$$

$$\frac{x_1}{24+16} = \frac{-x_2}{28+12} = \frac{x_3}{56-36} = K_3 \Rightarrow \text{or } \frac{x_1}{40} = \frac{x_2}{-40} = \frac{x_3}{20} = K_3$$

or $\frac{x_1}{2} = \frac{x_2}{-2} = \frac{x_3}{1} = K_1$

$$x_1 = K_3 \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

Q.3 Verify Cayley Hamilton Theorem of the matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$

Sol^n $|A - \lambda I| = 0$

$$A - \lambda I = \begin{bmatrix} -\lambda & 1 & 2 \\ 1 & 2-\lambda & 3 \\ 3 & 1 & 1-\lambda \end{bmatrix}$$

Takin the determinant :-

$$|A - \lambda I| = -\lambda \begin{vmatrix} 2-\lambda & 3 \\ 1 & 1-\lambda \end{vmatrix} - 1 \begin{vmatrix} 1 & 3 \\ 3 & 1-\lambda \end{vmatrix} + 2 \begin{vmatrix} 1 & 2-\lambda \\ 3 & 1 \end{vmatrix}$$

$\Rightarrow \lambda^3 - 3\lambda^2 - 11\lambda - 8 = 0$ { Characteristic eqⁿ }

Replace λ by A :- $A^3 - 3A^2 - 11A - 8I = 0$

Compute Required Powers

$A^2 = A.A \Rightarrow \begin{bmatrix} 7 & 4 & 5 \\ 11 & 8 & 11 \\ 4 & 6 & 10 \end{bmatrix}$, $A^3 = A^2.A \Rightarrow \begin{bmatrix} 19 & 20 & 31 \\ 41 & 38 & 57 \\ 36 & 26 & 36 \end{bmatrix}$

Substitute in the equation, we get

$\Rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Hence, the given matrix satisfies its own characteristic equation.

PART - C

Q. 1 Find the eigen values and eigen vector of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

Solⁿ Eigen value $= |A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix}$

$\Rightarrow (6-\lambda)(9 - 6\lambda + \lambda^2) + 2(-6 + 2\lambda + 2) + 2(2 - 6 + 2\lambda) = 0$

$\Rightarrow \lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$

$\Rightarrow (\lambda - 8)(\lambda - 2)^2 = 0$

$\lambda = 2, 2, 8$

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Eigenvectors :- Let x be eigen vector, $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ + $(A-\lambda I)X = 0$

$$\begin{bmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{——— (1)}$$

Put $\lambda = 8$ in eq^n (1), then
$$-2x_1 - 2x_2 + 2x_3 = 0$$
$$-2x_1 - 5x_2 - x_3 = 0$$
$$2x_1 - x_2 - 5x_3 = 0$$

Solving the system gives :

$$x = -y = z$$

$$x_1 = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$$

Put $\lambda = 2$ in eq^n (1), then
$$4x_1 - 2x_2 + 2x_3 = 0$$
$$-2x_1 + x_2 - x_3 = 0$$
$$2x_1 - x_2 + x_3 = 0$$

Solving the system gives :

$$x_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad x_3 = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

Eigen Values = 8, 2, 2 , Eigen Vector = (-1, 1, -1), (1, 2, 0), (-1, 0, 2)

Q.2 Verify Cayley Hamilton Theorem and hence find A^{-1} of the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & 2 \end{bmatrix}$$

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Sol^n Characteristic Equation $\Rightarrow$ $|A-\lambda I| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & -2 & 2 \\ 1 & 2-\lambda & 3 \\ 0 & -1 & 2-\lambda \end{vmatrix} = 0$

$\Rightarrow (1-\lambda)(4-4\lambda+\lambda^2) + 2(2-\lambda) + 2(-1)$

$\Rightarrow \lambda^3 - 5\lambda^2 + 13\lambda - 9 = 0$

Replace λ by A: $A^3 - 5A^2 + 13A - 9I = 0$

Hence Verified (Since every square matrix satisfies its characteristic equation).

Multiply by A^{-1} $\Rightarrow$ $A^2 - 5A + 13I - 9A^{-1} = 0$

$9A^{-1} = A^2 - 5A + 13I$

$A^{-1} = \frac{1}{9}(A^2 - 5A + 13I)$

Compute : $A^2 = \begin{bmatrix} -1 & -8 & 0 \\ 3 & -1 & 14 \\ -1 & -4 & 1 \end{bmatrix}$

Substitute $A^2 - 5A + 13I$ $\Rightarrow$ $\cancel{\begin{bmatrix} 5 & 2 & -10 \\ -2 & 0 & -1 \\ -1 & 1 & 2 \end{bmatrix}}$

$\Rightarrow \begin{bmatrix} 7 & 2 & -10 \\ -2 & 2 & -1 \\ -1 & 1 & 4 \end{bmatrix}$

$A^{-1} = \frac{1}{9}\begin{bmatrix} 7 & 2 & -10 \\ -2 & 2 & -1 \\ -1 & 1 & 4 \end{bmatrix}$

Q.3 Diagonalize the matrix $A = \begin{bmatrix} 2 & 2 & -7 \\ 2 & 1 & 2 \\ 0 & 1 & -3 \end{bmatrix}$

Sol^n Eigen Value $= |A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 2-\lambda & 2 & -7 \\ 2 & 1-\lambda & 2 \\ 0 & 1 & -3-\lambda \end{vmatrix} = 0$ $\quad D = P^{-1}AP$

$\Rightarrow (2-\lambda)(-3+3\lambda-\lambda+\lambda^2-2) - 2(-6-2\lambda) - 7(2) = 0$

$\Rightarrow -10 + 4\lambda + 2\lambda^2 + 5\lambda - 2\lambda^2 - \lambda^3 + 12 + 4\lambda - 14 = 0$

$-\lambda^3 + 13\lambda - 12 = 0$

$\lambda^3 - 13\lambda + 12 = 0$

$\lambda = 1, 3, -4$

~~Eigen Values = $(A - \lambda I)X = 0$~~, Putting $\lambda = 1$ / $\lambda = 3$ / $\lambda = -4$

$X_1 = \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix}$ / $X_2 = \begin{bmatrix} 5 \\ 6 \\ 1 \end{bmatrix}$ / $X_3 = \begin{bmatrix} 3 \\ -2 \\ 2 \end{bmatrix}$

So P matrix $= \begin{bmatrix} -1 & 5 & 3 \\ 4 & 6 & -2 \\ 1 & 1 & 2 \end{bmatrix}$

Now $D = PAP^{-1}$

$D = \begin{bmatrix} -1 & 5 & 3 \\ 4 & 6 & -2 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 & -7 \\ 2 & 1 & 2 \\ 0 & 1 & -3 \end{bmatrix} \begin{bmatrix} -1/5 & 1/10 & -14/35 \\ 1/7 & 1/14 & -1/7 \\ 1/35 & -3/35 & 13/35 \end{bmatrix}$

$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -4 \end{bmatrix}$

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